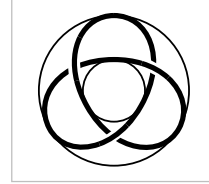


The relation between the fundamental group of a product space and the fundamental groups of its factors is as simple as one could wish:

Proposition 1.12. $\pi_1(X \times Y)$ is isomorphic to $\pi_1(X) \times \pi_1(Y)$ if X and Y are path-connected.

Proof: A basic property of the product topology is that a map $f: Z \rightarrow X \times Y$ is continuous iff the maps $g: Z \rightarrow X$ and $h: Z \rightarrow Y$ defined by $f(z) = (g(z), h(z))$ are both continuous. Hence a loop f in $X \times Y$ based at (x_0, y_0) is equivalent to a pair of loops g in X and h in Y based at x_0 and y_0 respectively. Similarly, a homotopy f_t of a loop in $X \times Y$ is equivalent to a pair of homotopies g_t and h_t of the corresponding loops in X and Y . Thus we obtain a bijection $\pi_1(X \times Y, (x_0, y_0)) \approx \pi_1(X, x_0) \times \pi_1(Y, y_0)$, $[f] \mapsto ([g], [h])$. This is obviously a group homomorphism, and hence an isomorphism. $\square$

Example 1.13: The Torus. By the proposition we have an isomorphism $\pi_1(S^1 \times S^1) \approx \mathbb{Z} \times \mathbb{Z}$. Under this isomorphism a pair $(p, q) \in \mathbb{Z} \times \mathbb{Z}$ corresponds to a loop that winds p times around one S^1 factor of the torus and q times around the other S^1 factor, for example the loop $\omega_{pq}(s) = (\omega_p(s), \omega_q(s))$. Interestingly, this loop can be knotted, as the figure shows for the case $p = 3, q = 2$. The knots that arise in this fashion, the so-called *torus knots*, are studied in Example 1.24.



More generally, the n -dimensional torus, which is the product of n circles, has fundamental group isomorphic to the product of n copies of $\mathbb{Z}$. This follows by induction on n .

Induced Homomorphisms

Suppose $\varphi: X \rightarrow Y$ is a map taking the basepoint $x_0 \in X$ to the basepoint $y_0 \in Y$. For brevity we write $\varphi: (X, x_0) \rightarrow (Y, y_0)$ in this situation. Then φ induces a homomorphism $\varphi_*: \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$, defined by composing loops $f: I \rightarrow X$ based at x_0 with φ , that is, $\varphi_*[f] = [\varphi f]$. This induced map φ_* is well-defined since a homotopy f_t of loops based at x_0 yields a composed homotopy φf_t of loops based at y_0 , so $\varphi_*[f_0] = [\varphi f_0] = [\varphi f_1] = \varphi_*[f_1]$. Furthermore, φ_* is a homomorphism since $\varphi(f \cdot g) = (\varphi f) \cdot (\varphi g)$, both functions having the value $\varphi f(2s)$ for $0 \leq s \leq 1/2$ and the value $\varphi g(2s - 1)$ for $1/2 \leq s \leq 1$.

Two basic properties of induced homomorphisms are:

- $(\varphi\psi)_* = \varphi_*\psi_*$ for a composition $(X, x_0) \xrightarrow{\psi} (Y, y_0) \xrightarrow{\varphi} (Z, z_0)$.
- $\mathbb{1}_* = \mathbb{1}$, which is a concise way of saying that the identity map $\mathbb{1}: X \rightarrow X$ induces the identity map $\mathbb{1}: \pi_1(X, x_0) \rightarrow \pi_1(X, x_0)$.

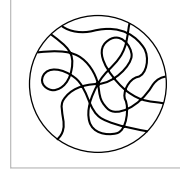
The first of these follows from the fact that composition of maps is associative, so $(\varphi\psi)f = \varphi(\psi f)$, and the second is obvious. These two properties of induced homomorphisms are what makes the fundamental group a functor. The formal definition

of a functor requires the introduction of certain other preliminary concepts, however, so we postpone this until it is needed in §2.3.

If φ is a homeomorphism with inverse ψ then φ_* is an isomorphism with inverse ψ_* since $\varphi_*\psi_* = (\varphi\psi)_* = \mathbb{1}_* = \mathbb{1}$ and similarly $\psi_*\varphi_* = \mathbb{1}$. We will use this fact in the following calculation of the fundamental groups of higher-dimensional spheres:

|| **Proposition 1.14.** $\pi_1(S^n) = 0$ if $n \geq 2$.

Proof: Let f be a loop in S^n at a chosen basepoint x_0 . If the image of f is disjoint from some other point $x \in S^n$ then f is nullhomotopic since $S^n - \{x\}$ is homeomorphic to $\mathbb{R}^n$, which is simply-connected. So it will suffice to homotope f to be nonsurjective. To do this we will look at a small open ball B in S^n about any point $x \neq x_0$ and see that the number of times that f enters B , passes through x , and leaves B is finite, and each of these portions of f can be pushed off x without changing the rest of f . At first glance this might appear to be a difficult task to achieve since the parts of f in B could be quite complicated geometrically, for example space-filling curves. But in fact it turns out to be rather easy.



The set $f^{-1}(B)$ is open in $(0, 1)$, hence is the union of a possibly infinite collection of disjoint open intervals (a_i, b_i) . The compact set $f^{-1}(x)$ is contained in the union of these intervals, so it must be contained in the union of finitely many of them. Consider one of the intervals (a_i, b_i) meeting $f^{-1}(x)$. The path f_i obtained by restricting f to the closed interval $[a_i, b_i]$ lies in the closure of B , and its endpoints $f(a_i)$ and $f(b_i)$ lie in the boundary of B . If $n \geq 2$, we can choose a path g_i from $f(a_i)$ to $f(b_i)$ in the closure of B but disjoint from x . For example, we could choose g_i to lie in the boundary of B , which is a sphere of dimension $n - 1$, hence path-connected if $n \geq 2$. Since the closure of B is homeomorphic to a convex set in $\mathbb{R}^n$ and hence simply-connected, the path f_i is homotopic to g_i by Proposition 1.6, so we may homotope f by deforming f_i to g_i . After repeating this process for each of the intervals (a_i, b_i) that meet $f^{-1}(x)$, we obtain a loop g homotopic to the original f and with $g(I)$ disjoint from x . $\square$

Example 1.15. For a point x in $\mathbb{R}^n$, the complement $\mathbb{R}^n - \{x\}$ is homeomorphic to $S^{n-1} \times \mathbb{R}$, so by Proposition 1.12 $\pi_1(\mathbb{R}^n - \{x\})$ is isomorphic to $\pi_1(S^{n-1}) \times \pi_1(\mathbb{R})$. Hence $\pi_1(\mathbb{R}^n - \{x\})$ is $\mathbb{Z}$ for $n = 2$ and trivial for $n > 2$.

Here is an application of this calculation:

|| **Corollary 1.16.** $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$.

Proof: Suppose $f: \mathbb{R}^2 \rightarrow \mathbb{R}^n$ is a homeomorphism. The case $n = 1$ is easily disposed of since $\mathbb{R}^2 - \{0\}$ is path-connected but the homeomorphic space $\mathbb{R}^n - \{f(0)\}$ is not path-connected when $n = 1$. When $n > 2$ we cannot distinguish $\mathbb{R}^2 - \{0\}$ from

$\mathbb{R}^n - \{f(0)\}$ by the number of path-components, but by the preceding calculation of $\pi_1(\mathbb{R}^n - \{x\})$ we can distinguish them by their fundamental groups. $\square$

The more general statement that $\mathbb{R}^m$ is not homeomorphic to $\mathbb{R}^n$ if $m \neq n$ can be proved in the same way using either the higher homotopy groups or homology groups. In fact, nonempty open sets in $\mathbb{R}^m$ and $\mathbb{R}^n$ can be homeomorphic only if $m = n$, as we will show in Theorem 2.19 using homology.

Induced homomorphisms allow relations between spaces to be transformed into relations between their fundamental groups. Here is an illustration of this principle:

Proposition 1.17. *If a space X retracts onto a subspace A , then the homomorphism $i_* : \pi_1(A, x_0) \rightarrow \pi_1(X, x_0)$ induced by the inclusion $i : A \hookrightarrow X$ is injective. If A is a deformation retract of X , then i_* is an isomorphism.*

Proof: If $r : X \rightarrow A$ is a retraction, then $ri = \mathbb{1}$, hence $r_*i_* = \mathbb{1}$, which implies that i_* is injective. If $r_t : X \rightarrow X$ is a deformation retraction of X onto A , so $r_0 = \mathbb{1}$, $r_t|_A = \mathbb{1}$, and $r_1(X) \subset A$, then for any loop $f : I \rightarrow X$ based at $x_0 \in A$ the composition r_tf gives a homotopy of f to a loop in A , so i_* is also surjective. $\square$

This gives another way of seeing that S^1 is not a retract of D^2 , a fact we showed earlier in the proof of the Brouwer fixed point theorem, since the inclusion-induced map $\pi_1(S^1) \rightarrow \pi_1(D^2)$ is a homomorphism $\mathbb{Z} \rightarrow 0$ that cannot be injective.

The exact group-theoretic analog of a retraction is a homomorphism ρ of a group G onto a subgroup H such that ρ restricts to the identity on H . In the notation above, if we identify $\pi_1(A)$ with its image under i_* , then r_* is such a homomorphism from $\pi_1(X)$ onto the subgroup $\pi_1(A)$. The existence of a retracting homomorphism $\rho : G \rightarrow H$ is quite a strong condition on H . If H is a normal subgroup, it implies that G is the direct product of H and the kernel of ρ . If H is not normal, then G is what is called in group theory the semi-direct product of H and the kernel of ρ .

Recall from Chapter 0 the general definition of a homotopy as a family $\varphi_t : X \rightarrow Y$, $t \in I$, such that the associated map $\Phi : X \times I \rightarrow Y$, $\Phi(x, t) = \varphi_t(x)$, is continuous. If φ_t takes a subspace $A \subset X$ to a subspace $B \subset Y$ for all t , then we speak of a homotopy of maps of pairs, $\varphi_t : (X, A) \rightarrow (Y, B)$. In particular, a **basepoint-preserving homotopy** $\varphi_t : (X, x_0) \rightarrow (Y, y_0)$ is the case that $\varphi_t(x_0) = y_0$ for all t . Another basic property of induced homomorphisms is their invariance under such homotopies:

- If $\varphi_t : (X, x_0) \rightarrow (Y, y_0)$ is a basepoint-preserving homotopy, then $\varphi_{0*} = \varphi_{1*}$.

This holds since $\varphi_{0*}[f] = [\varphi_0 f] = [\varphi_1 f] = \varphi_{1*}[f]$, the middle equality coming from the homotopy $\varphi_t f$.

There is a notion of homotopy equivalence for spaces with basepoints. One says $(X, x_0) \simeq (Y, y_0)$ if there are maps $\varphi : (X, x_0) \rightarrow (Y, y_0)$ and $\psi : (Y, y_0) \rightarrow (X, x_0)$

with homotopies $\varphi\psi \simeq \mathbb{1}$ and $\psi\varphi \simeq \mathbb{1}$ through maps fixing the basepoints. In this case the induced maps on π_1 satisfy $\varphi_*\psi_* = (\varphi\psi)_* = \mathbb{1}_* = \mathbb{1}$ and likewise $\psi_*\varphi_* = \mathbb{1}$, so φ_* and ψ_* are inverse isomorphisms $\pi_1(X, x_0) \approx \pi_1(Y, y_0)$. This somewhat formal argument gives another proof that a deformation retraction induces an isomorphism on fundamental groups, since if X deformation retracts onto A then $(X, x_0) \simeq (A, x_0)$ for any choice of basepoint $x_0 \in A$.

Having to pay so much attention to basepoints when dealing with the fundamental group is something of a nuisance. For homotopy equivalences one does not have to be quite so careful, as the conditions on basepoints can actually be dropped:

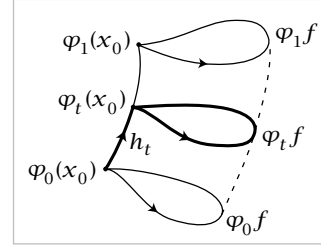
Proposition 1.18. *If $\varphi: X \rightarrow Y$ is a homotopy equivalence, then the induced homomorphism $\varphi_*: \pi_1(X, x_0) \rightarrow \pi_1(Y, \varphi(x_0))$ is an isomorphism for all $x_0 \in X$.*

The proof will use a simple fact about homotopies that do not fix the basepoint:

Lemma 1.19. *If $\varphi_t: X \rightarrow Y$ is a homotopy and h is the path $\varphi_t(x_0)$ formed by the images of a basepoint $x_0 \in X$, then the three maps in the diagram at the right satisfy $\varphi_{0*} = \beta_h \varphi_{1*}$.*

$$\begin{array}{ccc} & \varphi_{1*} & \pi_1(Y, \varphi_1(x_0)) \\ \pi_1(X, x_0) & \searrow & \downarrow \beta_h \\ & \varphi_{0*} & \pi_1(Y, \varphi_0(x_0)) \end{array}$$

Proof: Let h_t be the restriction of h to the interval $[0, t]$, with a reparametrization so that the domain of h_t is still $[0, 1]$. Explicitly, we can take $h_t(s) = h(ts)$. Then if f is a loop in X at the basepoint x_0 , the product $h_t \cdot (\varphi_t f) \cdot \bar{h}_t$ gives a homotopy of loops at $\varphi_0(x_0)$. Restricting this homotopy to $t = 0$ and $t = 1$, we see that $\varphi_{0*}([f]) = \beta_h(\varphi_{1*}([f]))$. $\square$



Proof of 1.18: Let $\psi: Y \rightarrow X$ be a homotopy-inverse for φ , so that $\varphi\psi \simeq \mathbb{1}$ and $\psi\varphi \simeq \mathbb{1}$. Consider the maps

$$\pi_1(X, x_0) \xrightarrow{\varphi_*} \pi_1(Y, \varphi(x_0)) \xrightarrow{\psi_*} \pi_1(X, \psi\varphi(x_0)) \xrightarrow{\varphi_*} \pi_1(Y, \varphi\psi\varphi(x_0))$$

The composition of the first two maps is an isomorphism since $\psi\varphi \simeq \mathbb{1}$ implies that $\psi_*\varphi_* = \beta_h$ for some h , by the lemma. In particular, since $\psi_*\varphi_*$ is an isomorphism, φ_* is injective. The same reasoning with the second and third maps shows that ψ_* is injective. Thus the first two of the three maps are injections and their composition is an isomorphism, so the first map φ_* must be surjective as well as injective. $\square$

Exercises

1. Show that composition of paths satisfies the following cancellation property: If $f_0 \cdot g_0 \simeq f_1 \cdot g_1$ and $g_0 \simeq g_1$ then $f_0 \simeq f_1$.
2. Show that the change-of-basepoint homomorphism β_h depends only on the homotopy class of h .
3. For a path-connected space X , show that $\pi_1(X)$ is abelian iff all basepoint-change homomorphisms β_h depend only on the endpoints of the path h .
4. A subspace $X \subset \mathbb{R}^n$ is said to be *star-shaped* if there is a point $x_0 \in X$ such that, for each $x \in X$, the line segment from x_0 to x lies in X . Show that if a subspace $X \subset \mathbb{R}^n$ is locally star-shaped, in the sense that every point of X has a star-shaped neighborhood in X , then every path in X is homotopic in X to a piecewise linear path, that is, a path consisting of a finite number of straight line segments traversed at constant speed. Show this applies in particular when X is open or when X is a union of finitely many closed convex sets.
5. Show that for a space X , the following three conditions are equivalent:
 - (a) Every map $S^1 \rightarrow X$ is homotopic to a constant map, with image a point.
 - (b) Every map $S^1 \rightarrow X$ extends to a map $D^2 \rightarrow X$.
 - (c) $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.

Deduce that a space X is simply-connected iff all maps $S^1 \rightarrow X$ are homotopic. [In this problem, 'homotopic' means 'homotopic without regard to basepoints.']

6. We can regard $\pi_1(X, x_0)$ as the set of basepoint-preserving homotopy classes of maps $(S^1, s_0) \rightarrow (X, x_0)$. Let $[S^1, X]$ be the set of homotopy classes of maps $S^1 \rightarrow X$, with no conditions on basepoints. Thus there is a natural map $\Phi: \pi_1(X, x_0) \rightarrow [S^1, X]$ obtained by ignoring basepoints. Show that Φ is onto if X is path-connected, and that $\Phi([f]) = \Phi([g])$ iff $[f]$ and $[g]$ are conjugate in $\pi_1(X, x_0)$. Hence Φ induces a one-to-one correspondence between $[S^1, X]$ and the set of conjugacy classes in $\pi_1(X)$, when X is path-connected.

7. Define $f: S^1 \times I \rightarrow S^1 \times I$ by $f(\theta, s) = (\theta + 2\pi s, s)$, so f restricts to the identity on the two boundary circles of $S^1 \times I$. Show that f is homotopic to the identity by a homotopy f_t that is stationary on one of the boundary circles, but not by any homotopy f_t that is stationary on both boundary circles. [Consider what f does to the path $s \mapsto (\theta_0, s)$ for fixed $\theta_0 \in S^1$.]

8. Does the Borsuk-Ulam theorem hold for the torus? In other words, for every map $f: S^1 \times S^1 \rightarrow \mathbb{R}^2$ must there exist $(x, y) \in S^1 \times S^1$ such that $f(x, y) = f(-x, -y)$?

9. Let A_1, A_2, A_3 be compact sets in $\mathbb{R}^3$. Use the Borsuk-Ulam theorem to show that there is one plane $P \subset \mathbb{R}^3$ that simultaneously divides each A_i into two pieces of equal measure.